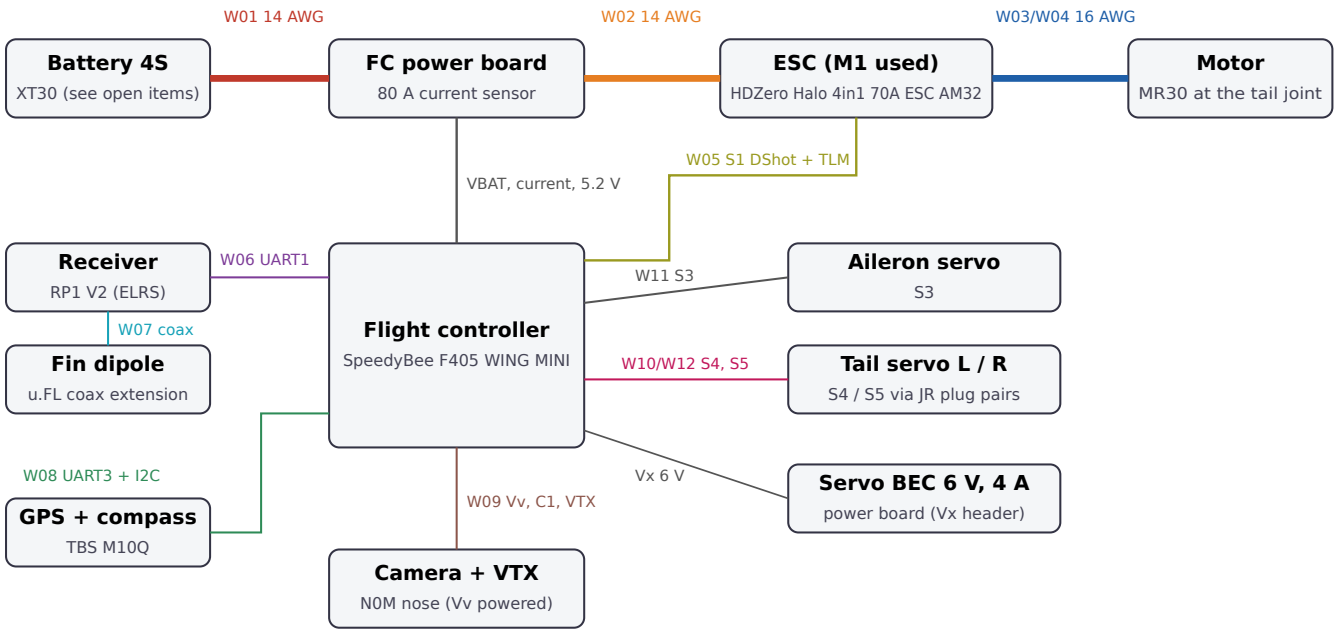
SkyG Speed 90B · Electrical integration manual

Power path, wire sizes and cut lengths, pinouts, routing, board settings, build order and tests. Cable lengths are measured on the 3D harness model of this build (the same routes the collision check uses). Generated from the design files; variant C6B, revision C6B-9 (2026-10-11), flight nose N0M.

> **Safety.** LiPo packs deliver over 100 A into a short. Work with the pack unplugged; first power-up through a smoke stopper or a current-limited supply; prop off for every bench test except the final static run; never charge inside the aircraft.

1. System overview



The battery feeds the flight controller's power board first, so its SpeedyBee 80 A current sensor measures everything; the power board feeds the ESC (HDZero Halo 4in1 70A ESC AM32, channel M1; M2 is the spare) and makes 5.2 V for the electronics and 6 V for the servos. The motor plugs in at the tail joint (MR30) so the tail module comes off. The receiver's antenna is a dipole inside the fin, clear of all metal.

2. Power budget and wire sizing

Cable	Circuit	Current A (pass / static)	Wire	Loop mm	R mOhm	Drop V	Loss W	dT K / pass	Connector rating
W01+W02	Battery to ESC (DC)	46 / 51	14 AWG	213	1.8	0.08 (0.6 %)	3.8	20	XT30: 15 A cont / 30 A peak (see open item)
W03+W04	Motor phases (per phase, RMS)	38 / 42	16 AWG	362	4.8	0.18 (1.2 %)	6.9	34	MR30: 30 A cont / 40 A for 1 min per contact

Test-base use (no governor, 70 % throttle cap): about 18 A. Currents from the propulsion model: 46.5 A at top speed (passes of <= 8 s), 51.3 A at full throttle on the ground. Phase RMS = sqrt(2/3) x DC. Temperature rise: copper only, no cooling, for one 8 s pass (an upper bound). Drop in % of the nominal 14.8 V.

Electronics: FC BEC 5.2 V / 2 A continuous (FC, GPS, receiver about 1.2 A together); servo BEC 6 V / 4 A continuous for three micro servos. Signal and servo wires: 26-28 AWG. Main DC path: 14 AWG (16 AWG would rise about 51 K per pass); motor phases: 16 AWG.

3. Cable schedule (cut list)

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W01	Battery lead	XT30 female (the pack plugs in), behind the pack under the GPS	FC power board BAT+ / BAT- pads	red BAT+, black BAT-	14	92	**125**	XT30 solder cups + 3/16 in heat shrink; pads tinned, 3 mm stripped	Over the ESC at z +2, down the 5 mm gap ahead of the FC stack.
W02	ESC feed	FC power board ESC+ / ESC- pads	HDZero Halo 4in1 70A ESC AM32 BAT+ / BAT- pads (XT60 lead removed)	red +, black -	14	14	**35**	Solder both ends; the ESC's low-ESR capacitor across its BAT pads (leads <= 10 mm)	Up the gap onto the ESC's rear end. 80 A current sensor in the path.
W03	Motor phases, ESC side	HDZero Halo 4in1 70A ESC AM32 M1 pads (A, B, C)	MR30 female (ESC side), low on the left ahead of the tail joint	3 x black (mark A/B/C)	16	81	**110**	Solder to the M1 pads; MR30 solder cups + 3 mm heat shrink	Up the gap and over the FC's left half (servo header row on the right).
W04	Motor phases, motor side	MR30 male	Motor phase wires (cut to 20 mm, butt-soldered + heat shrink)	3 x black	16	100	**140**	MR30 solder cups; butt splices with 3 mm heat shrink	Through the joint's upper quadrant, along the F3 top left of the fin dipole, down to the motor bulkhead. Unplug the MR30 before pulling the tail module.
W05	ESC signal	HDZero Halo 4in1 70A ESC AM32 8-pin SH1.0 lead (only M1, GND, TLM kept)	FC S1 header (signal, GND); TLM to the R4 pad	yellow M1 signal, black GND, white TLM	28	18	**45**	SH1.0 plug at the ESC (supplied lead, unused wires cut back and insulated); 1-pin Dupont housings at the FC header	Do NOT connect the lead's VBAT or CURR wires; never feed ESC voltage to the servo rail (Vx).
W06	Receiver lead	RadioMaster RP1 V2 pads (5V, GND, TX, RX), on the ESC	FC receiver header (5V, GND, RX1, TX1)	red 5V, black GND, yellow RP1-TX -> RX1, white RP1-RX -> TX1	28	22	**45**	Solder at the receiver; 4-pin Dupont housing at the FC	Low over the ESC, up the gap at y -5.5.

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W07	ELRS antenna coax	RP1 V2 u.FL	Dipole feed at the centre of the fin dipole (z 39.5)	1.13 mm coax	coax	241	**275**	Two pieces joined by a u.FL plug/jack 4 mm ahead of the tail joint: the receiver-side extension (u.FL male to u.FL female) and the T-antenna's own coax, which runs up the fin's groove beside the lower leg to its feed	Under the wing-screw bridges, right of the aileron hardware, over the tail servos, up the fin groove. Unplug the u.FL joint to remove the tail module. Antenna side (joint to feed) 92 mm: buy a 2.4 GHz T/dipole antenna with at least 110 mm of coax (the RP1's stock T-antenna is shorter), receiver side 148 mm.
W08	GPS lead	TBS M10Q (VCC, GND, TX, RX, SDA, SCL)	FC GPS interface with compass (4V5, GND, RX3, TX3, SDA, SCL)	6 x 28 AWG (the model shows the bundle as 4)	28	90	**130**	Supplied GPS plug; solder or GH plug at the FC	Under the GPS carrier; leave a slack loop so the hatch lifts off.
W09	Video trunk (N0M nose)	Nose joint plug (camera on the nose; VTX on the nose or, dome nose, under the hatch)	FC pads: Vv, G, C1 (camera in), VTX (video out)	red +, black GND, yellow camera video, white VTX video	28	212	**275**	4-pin JST plug pair at the nose joint; solder at the FC	Under the battery floor plate, up through the floor hole ahead of the F1/F2 bulkhead.
W10	Tail servo leads (L, R)	Emax ES9052MD in F3 (its stock lead)	JR plug pair lying flat beside the aileron servo	brown GND, red +6 V, orange signal	26	64	stock	Stock servo plug	Along the floor and through the tail joint; the plug pair separates the tail module. Needs a stock lead of at least the route length + 30 mm.
W12	Tail servo extensions (L, R)	JR plug pair (beside the aileron servo)	FC S4 (left) / S5 (right)	brown, red, orange	26	15	stock	100 mm stock extension (BOM line 35)	Along the floor to the FC's rear edge.

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W11	Aileron servo lead	Emax ES9052MD lead (stock), aileron servo in the wing bay	FC S3 header	brown, red, orange	26	60	stock	Stock plug	Under the floor beside the servo, up behind the FC.

Cut length = route length on the model x 1.10 (slack, 1.15 for plugged leads) + the termination allowances, rounded up to 5 mm. Total cut length of made cables: 1180 mm. Stock servo leads are used as supplied (W10 servo side, W11).

4. Connections and pinouts

SpeedyBee F405 WING MINI

Pad / header	Connects	Setting / note
BAT+ / BAT- (power board)	W01 battery lead	14 AWG red / black
ESC+ / ESC- (power board)	W02 ESC feed	14 AWG red / black
S1 header: signal, GND	W05 ESC M1 signal + GND (no + wire)	DShot300 bidirectional (SERVO1_FUNCTION 70)
R4 pad	W05 ESC TLM (optional)	AM32 serial telemetry, fallback rpm source
S3 header	W11 aileron servo	SERVO3_FUNCTION 4
S4 / S5 headers	W10 tail servos L / R (via extensions)	SERVO4/5_FUNCTION 79 / 80 (V-tail)
Receiver header: 5V, GND, RX1, TX1	W06 RP1 V2	CRSF, SERIAL1_PROTOCOL 23
GPS interface: 4V5, GND, RX3, TX3, SDA, SCL	W08 TBS M10Q	SERIAL3_PROTOCOL 5; compass on I2C
Vv, G, C1	W09 camera (power + video in)	Vv = battery voltage by default: the camera must accept 4S
Vv, G, VTX (T5 for HD)	W09 VTX (power + video out)	Analog OSD on the FC; HD/DJI: T5/R5 MSP DisplayPort
USB extender (UART6)	Wireless USB extender	Setup on the ground; not flown

HDZero Halo 4in1 70A ESC AM32: 8-pin SH1.0 lead

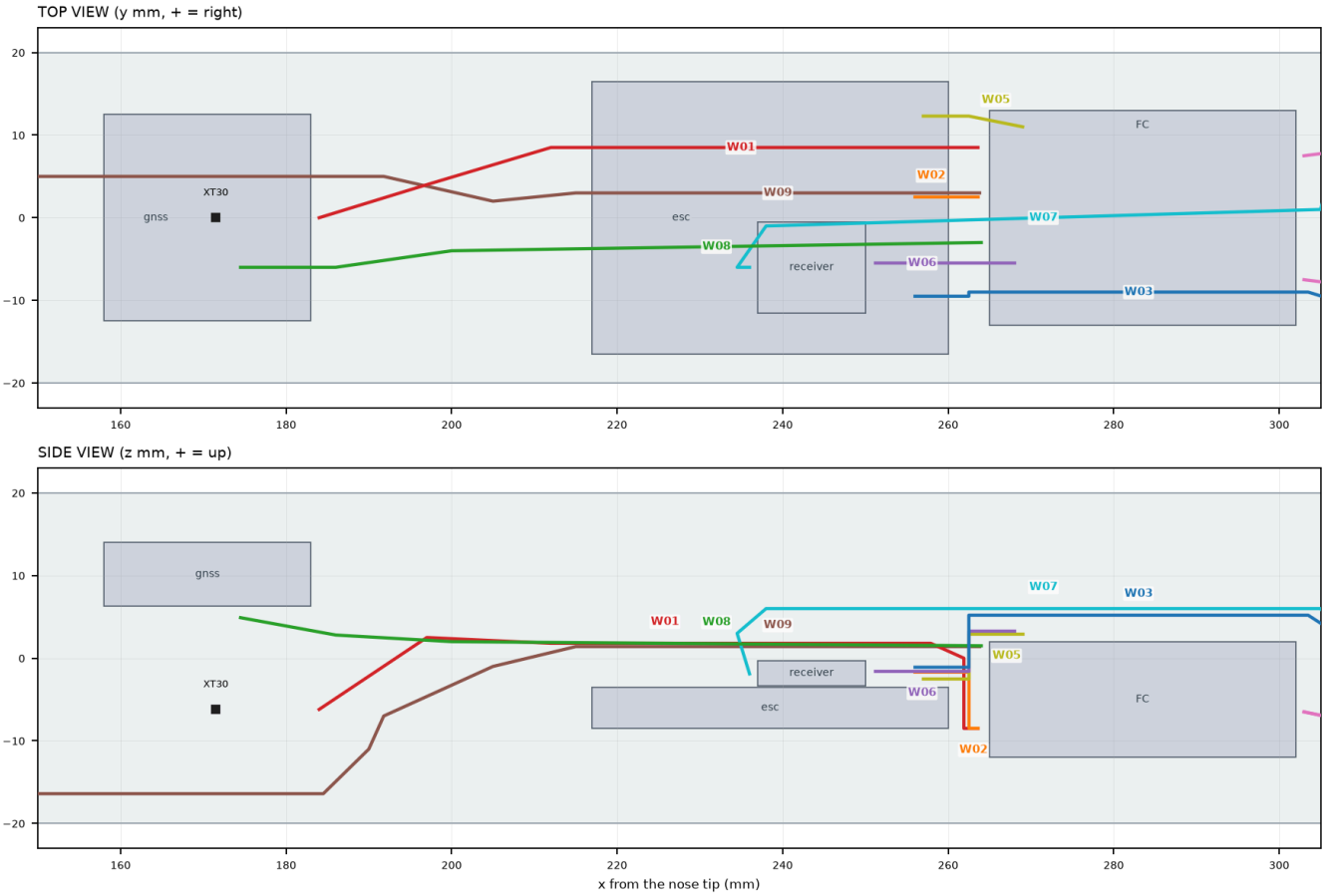
Pin	Wire	Use
M1 signal	yellow	W05 -> FC S1 signal
GND	black	W05 -> FC S1 GND
TLM	white	W05 -> FC R4 (optional)
VBAT	-	Cut back and insulate: the FC is powered by its own power board
CURR	-	Cut back and insulate: the FC's own 80 A sensor is used
M2, M3, M4	-	Cut back and insulate (unused channels; M2 is the spare)

> Never connect the ESC lead's VBAT pin to the FC's servo header (+) or any 5 V pad: it is battery voltage.

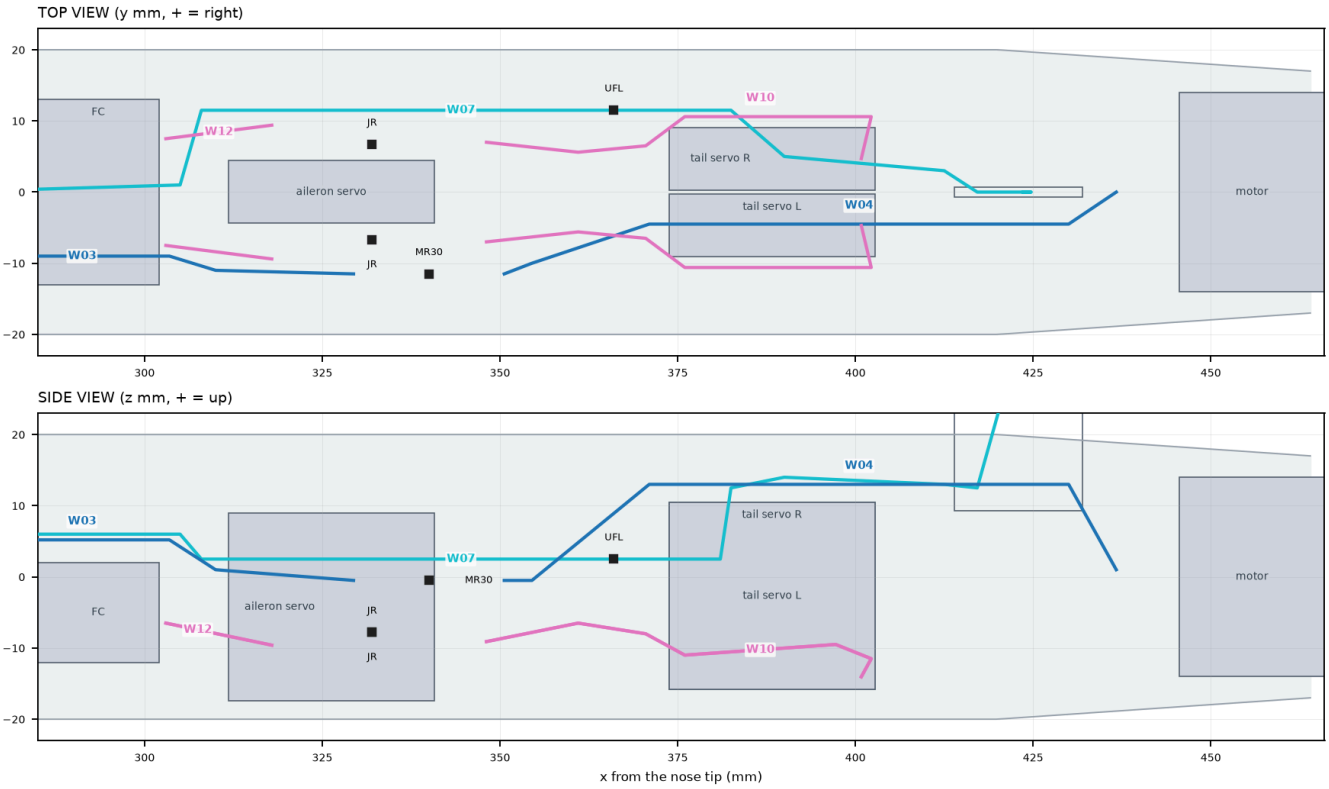
Connectors

Connector	Gender / pins	Note
XT30 (battery)	Female on the aircraft lead, male on the pack	Red +, black -; check polarity on the housing's chamfer
MR30 (motor)	Female on the ESC side, male on the motor side	Phase order free: set the direction in AM32, not by swapping wires
JR servo plug	Brown GND, red +6 V, orange signal	Same orientation on every header: signal towards the board edge marked S
u.FL (ELRS coax)	Plug on the receiver and at the tail joint	Max about 30 mating cycles; tape the joint, no tension on the coax

5. Routing



Front: battery bay, wing bay (ESC, FC, receiver).

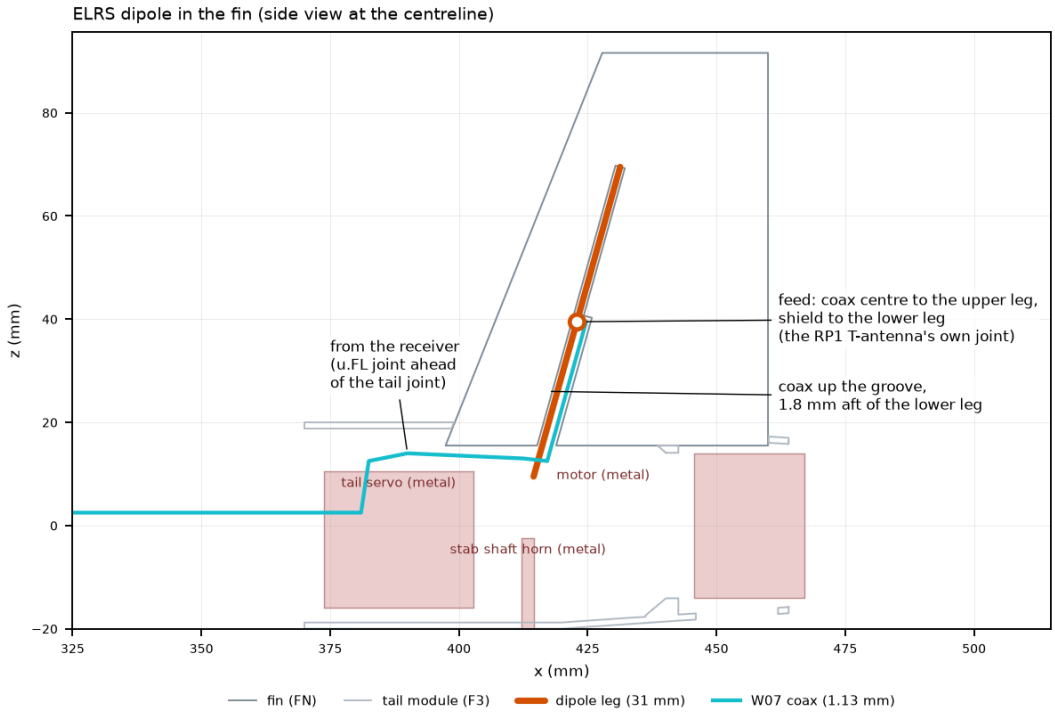


Rear: aileron servo, tail joint, tail module, fin. Cable routes from the 3D harness model (x from the nose tip); squares are connector pairs. Every route is collision-checked against every printed part, component and cable.

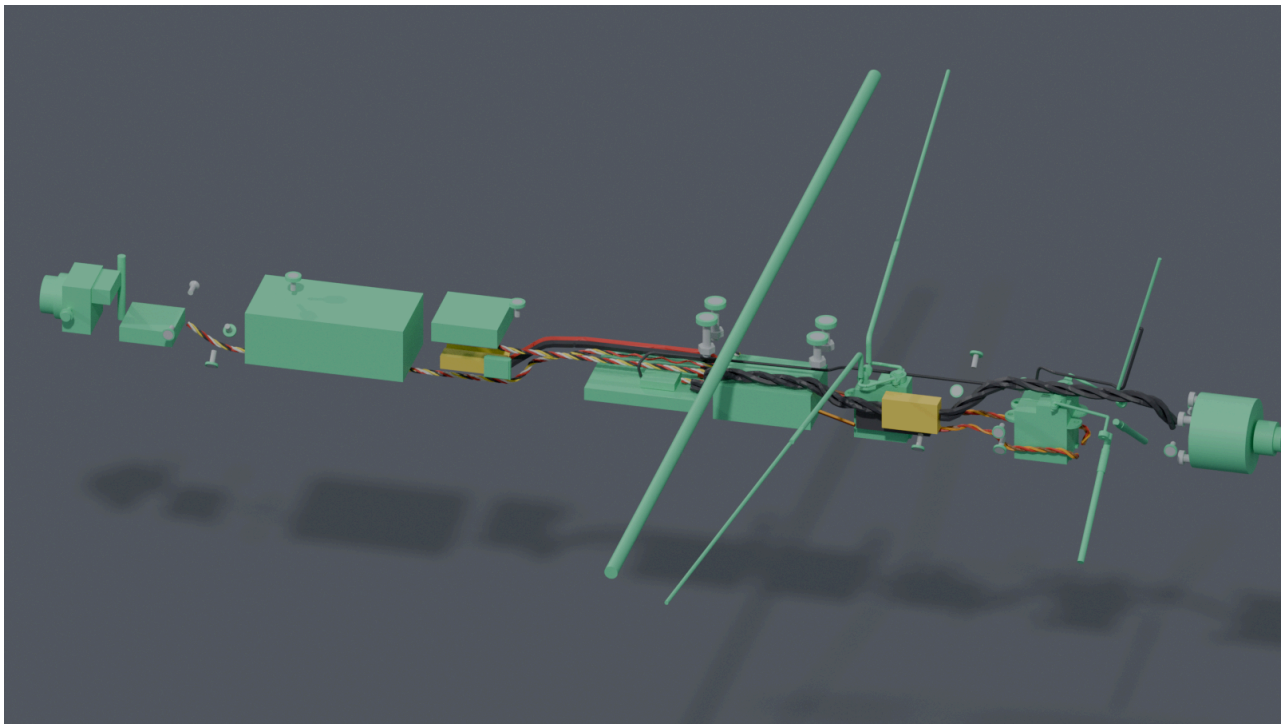


Render with the fin and tail module see-through: the dipole (green) and the antenna coax (cyan) entering the tail module from the u.FL joint, along the top and up the fin's groove beside the lower leg to the feed.

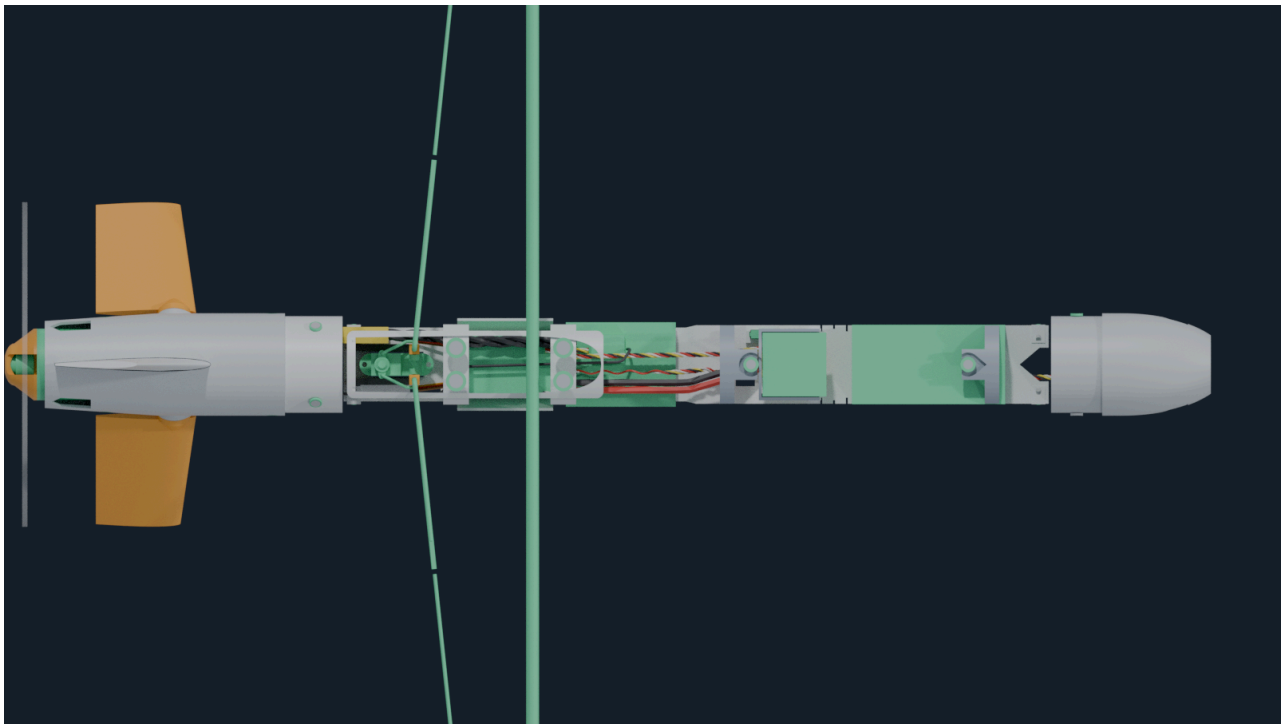
Receiver antenna in the fin



The RP1 V2 T-antenna straightened into a vertical dipole: two 31 mm legs in a groove on the fin's 30 % chord line, the feed (the T's joint) at the centre. Its own coax runs down a parallel groove beside the lower leg into the tail module and meets the 1.13 mm extension at the u.FL joint ahead of the tail joint. Clearance to the nearest metal (tail shaft): 12.9 mm. Fit: lay the antenna and coax in the grooves before the fin is glued into F3; three dots of epoxy, then tape over the grooves.



Render: harness with the fuselage hidden.



Render: top view, fuselage cut open.

Lanes in the 5 mm gap between the ESC and the FC stack (left to right, y in mm): motor phases W03 -9.5, receiver W06 -5.5, ESC feed W02 +2.5, battery W01 +8.5, ESC signal W05 +12.3. The video trunk W09 and the GPS lead W08 arrive over the ESC at z 0.5-1.5; the receiver coax W07 runs higher (z +6).

6. Board and ESC settings

Item	Setting	Why
Power board: servo BEC voltage jumper	6 V	Emax ES9052MD: 1.3 kgf cm at 6 V (1.1 at 5 V). Vx header = 6 V.
Power board: servo BEC supply jumper	Servo BEC (default)	Keep: the ESC has no BEC; never power Vx from the ESC lead.

Item	Setting	Why
Power board: Vv (VTX/camera) jumper	Battery voltage (default)	Only if camera and VTX accept 4S (13-17 V); otherwise 5 V (shares the servo BEC; then 6 V when the servo jumper is 6 V).
ESC (AM32): protocol	DShot300, bidirectional on	SERVO_BLH_MASK 1, SERVO_BLH_BDMASK 1, SERVO_BLH_OTYPE 5.
ESC (AM32): motor direction	Set on the bench	Pusher prop: the motor turns the prop so that it blows aft. Reverse in AM32 if wrong.
ESC (AM32): brake on stop	On	Holds the prop still for the belly landing.
ESC (AM32): motor poles	14	2207 motors are 12N14P: SERVO_BLH_POLES 14 for correct rpm.
FC: battery monitor	BATT_MONITOR 4, BATT_VOLT_MULT 11.0, BATT_AMP_PERVLT 50	Calibrate both against a meter (see the tests).
FC: current limit (until the governor)	THR_MAX per the ArduPilot document	Keeps the unloaded prop below its rating; full throttle bench runs <= 10 s (MR30, XT30 heating).

7. Build order

1. Bench, before anything goes into the fuselage: flash ArduPilot (SpeedyBeeF405WING), load the parameter file, set the power-board jumpers (table above).
2. ESC: unsolder the supplied XT60 lead. Tin the BAT pads and solder W02 (14 AWG, cut length in the schedule) and the low-ESR capacitor across them (short leads, capacitor laid flat beside the receiver position). Solder W03 to the M1 pads. Cut the SH1.0 lead's VBAT, CURR, M2-M4 wires back to 5 mm and insulate each.
3. FC: solder W01 and the free end of W02 to the BAT and ESC pads (14 AWG: 3 mm strip, pre-tin, 2-3 s per joint with a 60-80 W iron). Solder W08 (GPS), W09 (video) and the TLM wire of W05 (R4). Fit the 1-pin Dupont housings of W05 on S1 (signal + GND) and the 4-pin housing of W06 on the receiver header.
4. Continuity and insulation, unpowered: BAT+ to BAT- > 1 kOhm (capacitor charging is normal), no short between any motor phase and BAT-, polarity of the XT30 matches the pack.
5. Smoke stopper first power-up (a current-limited supply, 16 V / 0.5 A, or a smoke stopper on the pack): FC boots, ESC beeps, no heat anywhere. Then the pack directly.
6. Fit in F2, front to back: ESC on its foam tape (component side up), receiver on the ESC (left of centre), FC stack 5 mm behind the ESC (arrow forward, servo header row on the right). Route W01, W02, W05 and W06 down/up the 5 mm gap in their lanes (W01 at y +8.5, W02 at y +2.5, W05 at y +12.3, W06 at y -5.5).
7. Motor: W03 up the gap at y -9.5 and over the FC's left half to the MR30 low on the left ahead of the tail joint; W04 from the MR30 through the joint's upper quadrant and along the F3 top 4.5 mm left of centre to the motor.
8. Receiver antenna: W07 from the RP1 up over the bay at z +6 (under both wing-screw bridges), to the right side behind the FC, through the tail joint (u.FL joint here, taped), up over the tail servos to the fin dipole. The dipole: RP1 T-antenna straightened, 31 mm legs, in the fin's groove (30 % chord line), epoxy-tacked at three points.
9. Servos: aileron lead W11 under the floor to S3; tail servo leads through the joint to the JR plug pairs lying flat beside the aileron servo, extensions to S4 (left) and S5 (right).
10. Strain relief: Kapton tape at every lane change, nothing resting on the motor phases (W03/W04) or the ESC's FETs, a slack loop on W08 for the hatch, MR30 and the u.FL joint reachable through the tail joint.

8. Power-up and tests

Test	Procedure / pass criterion
Voltage	Read the pack voltage with a meter and in ArduPilot; adjust BATT_VOLT_MULT until they agree within 0.05 V.
Current	Run the motor (prop off) at a steady setting with a clamp meter or a watt meter on the pack lead; adjust BATT_AMP_PERVLT.
Motor direction and rpm	Prop off: short burst, check direction (pusher) and that ESC rpm appears in ArduPilot (bidirectional DShot).
Servos	Direction and travel of each surface against the ArduPilot document; no binding at full deflection; aileron reflex 3 deg TE up at neutral.
Radio and failsafe	Bind the RP1 V2; switch the transmitter off: RTL after 5 s. Range check at reduced power with the antenna in the fin.
GPS and compass	3D fix outdoors; compass calibration with the ESC and battery installed (current drawn near the compass).
Thermal	Static full throttle (prop on, aircraft held) for at most 10 s: ESC, wires, XT30 and MR30 hand-warm afterwards (connectors are near their ratings).
Video	Camera image and OSD (voltage, current, ground speed) in the goggles; VTX power and band legal for the event.

9. Open items (decide before ordering or flying)

- XT30 rating: Amass XT30 is rated 15 A continuous / 30 A peak. Flown as the test base (no governor, 70 % throttle cap) the pack current is about 18 A (prop law from the 51 A static figure): within the 30 A peak, so XT30 is acceptable for that use. With the governor fitted it would draw 46 A on a pass and 51 A at static full throttle, above the rating: then switch to XT60 (see below).
- XT30 rating: Amass XT30 is rated 15 A continuous / 30 A peak; this aircraft draws 46 A on a pass and 51 A at static full throttle. FPV builds often run XT30 this hard for seconds, but it is outside the rating. Options: an XT60 lead with ready-made packs that have XT60 (the connector pair needs about 16 x 8 mm behind the pack instead of 10 x 5 mm), or keep XT30 and limit full power to the short passes. Decide before ordering packs.
- Halo ESC thickness: modelled 5 mm with components. Measure on receipt; above 5.5 mm its top corners reach the F2 side-vent ducts.
- Halo 8-pin lead: the pin order is printed on the ESC; confirm M1, GND and TLM on the board before cutting the other wires.
- FC power-board pad positions: the routes end at the stack's front face; the cut lengths include 25-30 mm for reaching the pads wherever they sit on the board.

10. Tools and consumables

- Temperature-controlled iron 60-80 W with a chisel tip (14 AWG pads), fine tip for 28 AWG; leaded solder 0.5-0.8 mm; no-clean flux.
- Silicone wire: 14 AWG red + black, 16 AWG black (3 phases), 28 AWG in four colours; 1.13 mm u.FL extension.
- Heat shrink 1.5, 3, 4.8 mm; Kapton tape; double-sided foam tape; XT30 pair, MR30 pair, 1-pin and 4-pin Dupont housings with crimps.
- Multimeter, smoke stopper or current-limited bench supply, clamp or watt meter for the current calibration.